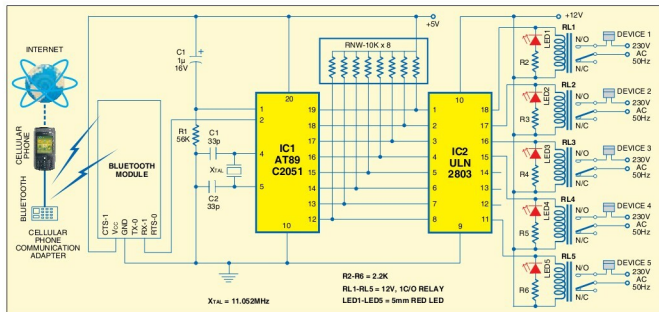




Circuit for home appliance control via Bluetooth-enabled cellular phone

absorbed. This change in the ultrasonic waves registers the position of the touch event and sends this information to the controller for processing.

The wave setup has no metallic layers on the screen, allowing for 100 per cent light throughput and perfect image clarity. This makes the surface acoustic wave system best for displaying detailed graphics (both the other systems have significant degradation in clarity). Surface-wave touchscreen panels are the most advanced of the three types, but these can be damaged by outside elements.

Another area in which the systems differ is the way stimuli is registered as a touch event. A resistive system registers a touch as long as the two layers make contact, which means that it doesn't matter if you touch it with your finger or a rubber ball. A capacitive system, on the other hand, must have a conductive input, usually your finger, in order to register a touch. The surface acoustic wave system works much like the resistive system, allowing a touch with almost any object—except hard and small objects like a pen tip.

Q3. How can we interface a bluetooth module with a microcontroller to control devices like door-

locks and lights from a cellphone?

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A3. The circuit of a simple cellphone-based device-control system using bluetooth communication link and microcontroller is shown above. It can be easily designed by engineering students and hobbyists. The advantage of bluetooth technology is that it doesn't require line-of-sight for signal transmission from the transmitter to the receiver.

This device control system overcomes the problems of manual switching (on/off) of different devices through bluetooth-enabled cellphone communicating with the bluetooth module interfaced with the microcontroller. It is a simple and cost-effective design to solve the problem of remote control for whatever reason it may be.

The circuit consists of a cellphone attached to a bluetooth adaptor (one could use a cellphone having inbuilt bluetooth), a bluetooth module interfaced with microcontroller AT89C2051, a driver IC ULN2803, five 12V relays, LEDs and associated discrete components.

At the heart of the circuit is microcontroller IC AT89C2051 (IC1). It is

a 20-pin, 8-bit microcontroller with 2 kB of Flash programmable and erasable read-only memory (PEROM), 128 bytes of RAM, 15 input/output lines, two 16-bit timers/counters, a five-vector two-level interrupt architecture, a duplex serial port, a precision analogue comparator, on-chip oscillator and clock circuitry.

The ULN2803 (IC2) consists of eight npn Darlington-connected transistors, which are ideally suitable for interfacing between low-logic-level digital circuitry and the higher current/voltage requirement circuits such as lamps, fans or other similar loads for a broad range of computer, industrial and consumer applications. It is used here for driving the devices through relay contacts. You can write a simple software program for the microcontroller of the bluetooth-enabled cellphone, employing simple coding and decoding of remote signals to perform switching of devices.

Answers compiled by Air Cmdr (Retd) P.D. Badoni

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